Exercice 6 :

AR(1) :

We know that $\text{Var}(\hat{r}_1) \simeq \frac{\phi^2}{n} = \frac{(0,5)^2}{100} = 0,0025$

$SE(\hat{r}_1) = 0,05$

the z score $= \frac{0,5}{0,05} = 10$

Conclusion : A result 10 standard deviation away from 0 is unusual.

MA(1) : Same goes for the MA(1), the result is unusual.